Prototype National Research Platform Classroom



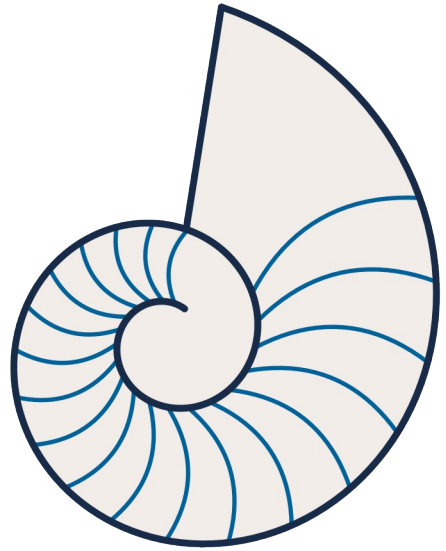
NAIRR AI Education Webinar Series

Daniel Diaz · San Diego Supercomputer Center, UC San Diego

Edge Webinar · NJEdge · 13 November 2026

Agenda

- What is NRP?
- Getting in: your login, a namespace, NAIRR Classroom
- What a course can use: CPUs, GPUs, storage, AI models
- JupyterHub for your class: what you control as the instructor
- Live demo: a class hub from one file
- Questions



Nautilus: the name of NRP's cluster

Slides, the demo notebook and the files:

https://training.nrp-nautilus.io/nairr-classroom

What NRP Is

What is NRP?

- One national **Kubernetes** cluster, **Nautilus**, for research and education
- Run by SDSC, Nebraska–Lincoln and MGHPCC; funded by NSF

https://dash.nrp-nautilus.io

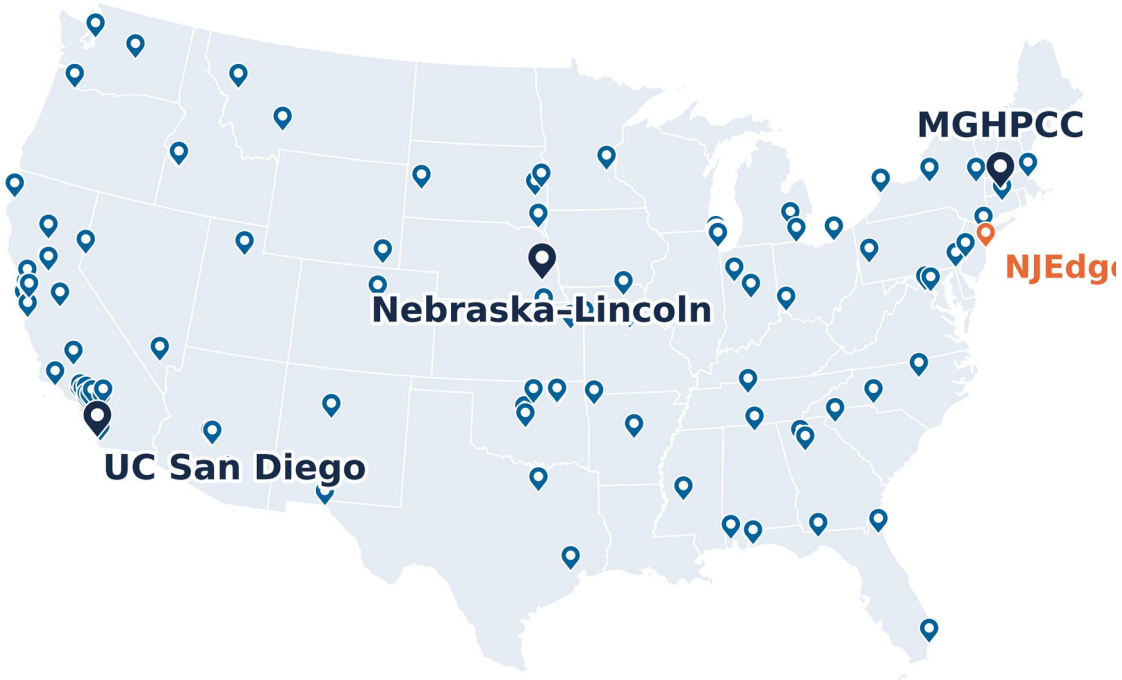


Nodes, GPUs, CPUs (hardware threads) and storage: cluster snapshot, 10 Nov 2026

What is NRP?

- One national **Kubernetes** cluster, **Nautilus**, for research and education
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- Hardware from 70+ institutions, on research and education networks
  - NJEdge hosts NRP nodes too (orange)
- Today:
  - 534 nodes at 129 sites
  - 1,460 GPUs in 33 models; 35,448 CPUs
  - 15 PB of raw Ceph storage

https://dash.nrp-nautilus.io



Sites from the NRP dashboard, 10 Nov 2026
Not drawn: 8 sites in Hawaii, Guam, Korea, Brazil and Europe

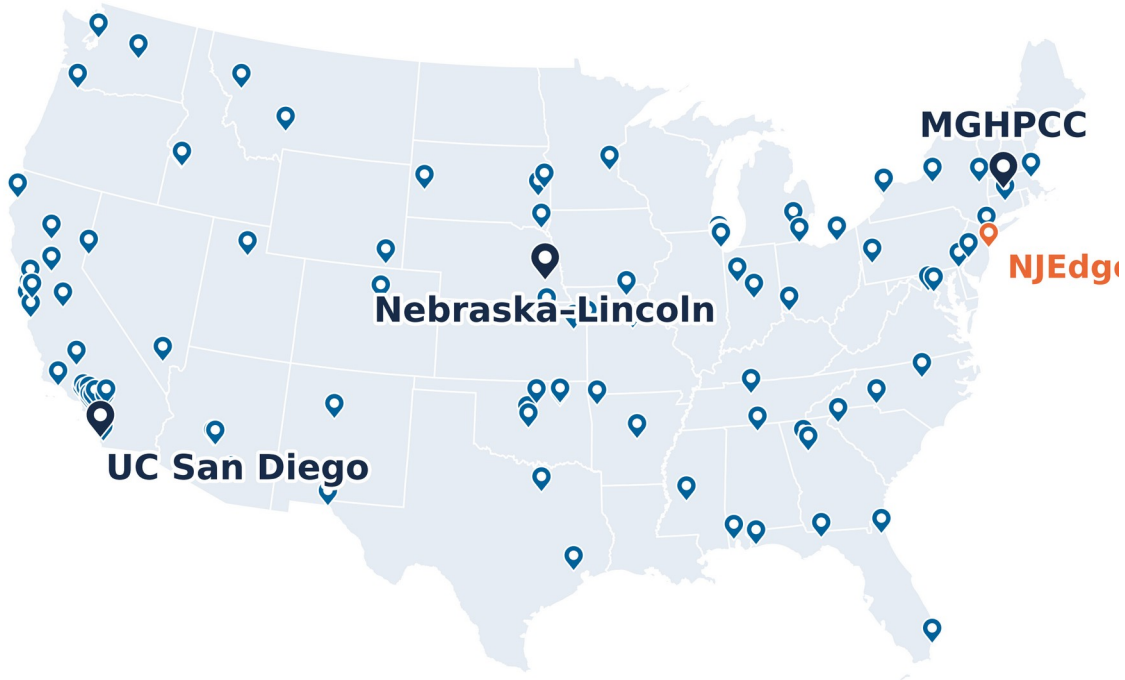
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- Today:
  - 534 nodes at 129 sites
  - 1,460 GPUs in 33 models; 35,448 CPUs
  - 15 PB of raw Ceph storage
- Free for US nonprofit institutions, community colleges included
- One team runs it from the OS up: no Kubernetes team needed on your campus

Nodes, GPUs, CPUs (hardware threads) and storage: cluster snapshot, 10 Nov 2026

https://dash.nrp-nautilus.io



Sites from the NRP dashboard, 10 Nov 2026
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From PRP to NRP

2015

Pacific Research Platform

Data transfer nodes (FIONAs) in campus Science DMZs, on CENIC and Pacific Wave

OAC-1541349

2018

Toward a National Research Platform

The same model, beyond the West Coast

OAC-1826967

2021

National Research Platform

An NSF Category II resource: GPUs, storage and services for research and education

OAC-2112167

2026

Today

One Kubernetes cluster, Nautilus: 534 nodes at 129 sites, and AI models as a service

Under each milestone: its NSF award.

NRP in Numbers

534
nodes

129
sites on the dashboard

1,460
GPUs in 33 models

35,448
CPUs (hardware threads)

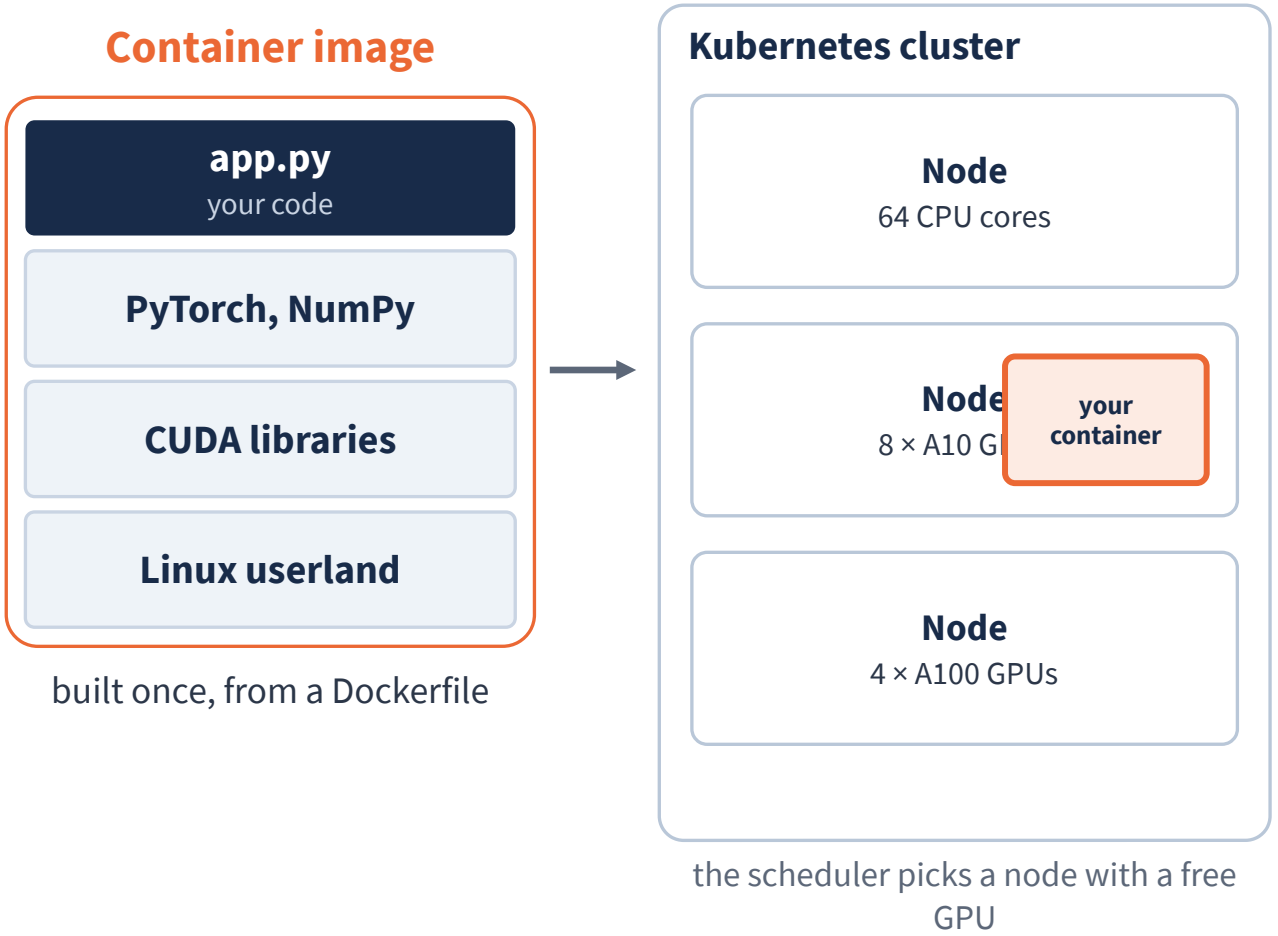
14.9 PB
raw Ceph in 9 clusters

2,334
namespaces: labs, courses, services

Cluster snapshot, 10 Nov 2026 (Kubernetes v1.34.11). Sites: the NRP dashboard.

Containers and Kubernetes

- **Container image:** an app and everything it needs, built once from a Dockerfile; runs the same anywhere
- **Kubernetes** runs containers across many machines: you declare what you want (an image, CPUs, memory, a GPU), the **scheduler** picks a **node** that has it, and controllers keep it that way
- A running container is a **pod**. Pods, Jobs, Deployments and Services live in a **namespace**: one per lab, course or service
- NRP is one Kubernetes cluster: 534 nodes



Getting In

Getting Access

1. Log in at **nrp.ai**

Log In → CILogon → your institution's own login

2. Get a namespace for your course

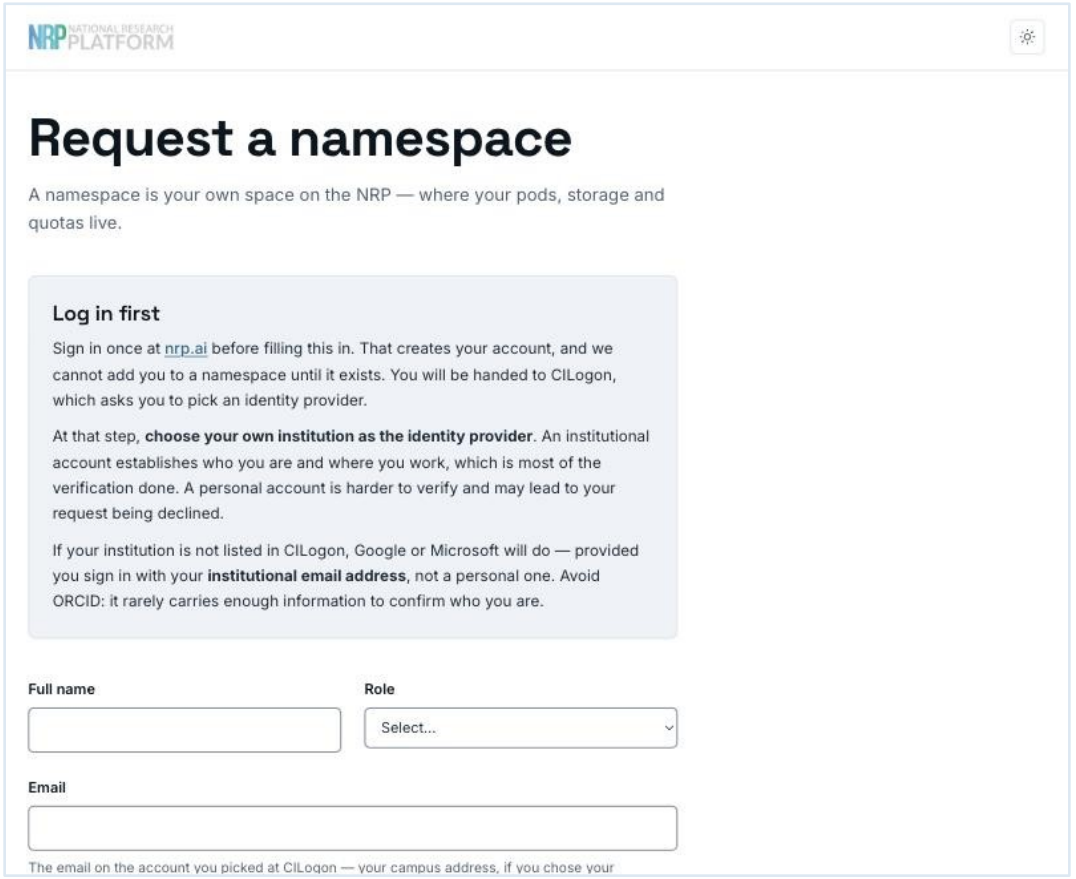
Request one at **get-started.nrp-nautilus.io**, or email the NRP team the course, the number of students and the start date

3. Bring your students

They log in the same way, with their campus account: no new passwords, nothing to install

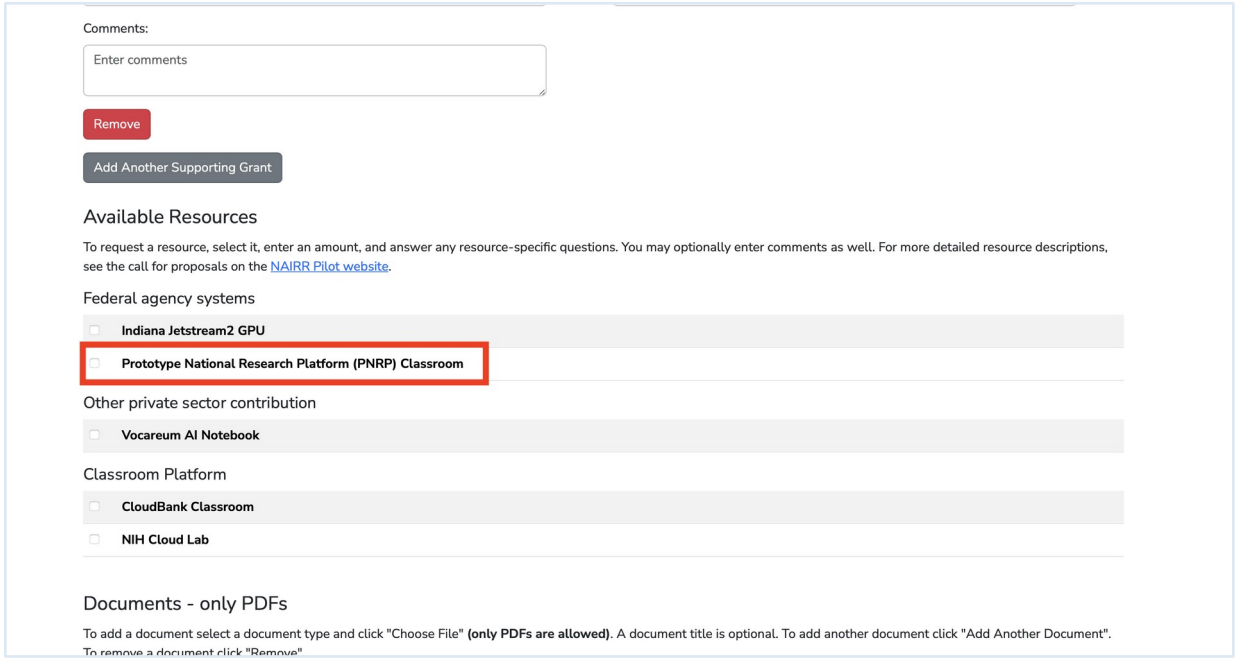
Who can request a namespace: faculty, staff, postdocs and research scientists. Free for US nonprofit institutions.

https://get-started.nrp-nautilus.io



NAIRR Classroom

- The **NAIRR Pilot** takes **Classroom** requests from US educators
- Your course, and what it needs, in **up to three pages**
- Courses that teach AI, in **any discipline**, for US-based students
- Reviewed as they arrive; **a decision within a month**
- Tick **Prototype NRP (PNRP) Classroom**: a Jupyter platform, CPUs, GPUs, storage and hosted AI models, run by the NRP team
- Not required to use NRP, but it gives your course priority



The NAIRR Classroom request form: NRP's option, among the other providers

https://nairrpilot.org

What is a Namespace?

- Your course's space on the cluster: one per course, lab or project
- You're its admin, and you add your TAs
- Your course hub, its storage and anything else you run lives in it
- Members see their own namespaces, not anyone else's
- 2,334 of them today

Nautilus cluster

bio-210
the instructor (admin), 2 TAs, 40 students · the course JupyterHub · shared datasets

ml-lab
a PI (admin) and three students · GPU jobs · a data volume · a web app

another-course
their members only · you can't see their servers or files

Ways In

JupyterHub
notebooks in the browser; NRP's, or your course's own
`jupyterhub-west.nrp-nautilus.io`

Coder
VS Code in the browser, for programming courses
`coder.nrp-nautilus.io`

kubectl
jobs, services and storage in your namespace
`nrp.ai/config`

GitLab
course repositories, and builds of course images
`gitlab.nrp-nautilus.io`

LLM API
hosted models for notebooks and apps
`ellm.nrp-nautilus.io/v1`

Chat
talk to the hosted models in a browser
`nrp-openwebui.nrp-nautilus.io`

Most students only ever see the first one. The rest are there when a course outgrows notebooks.

Resources for Your Course

What a Course Can Use

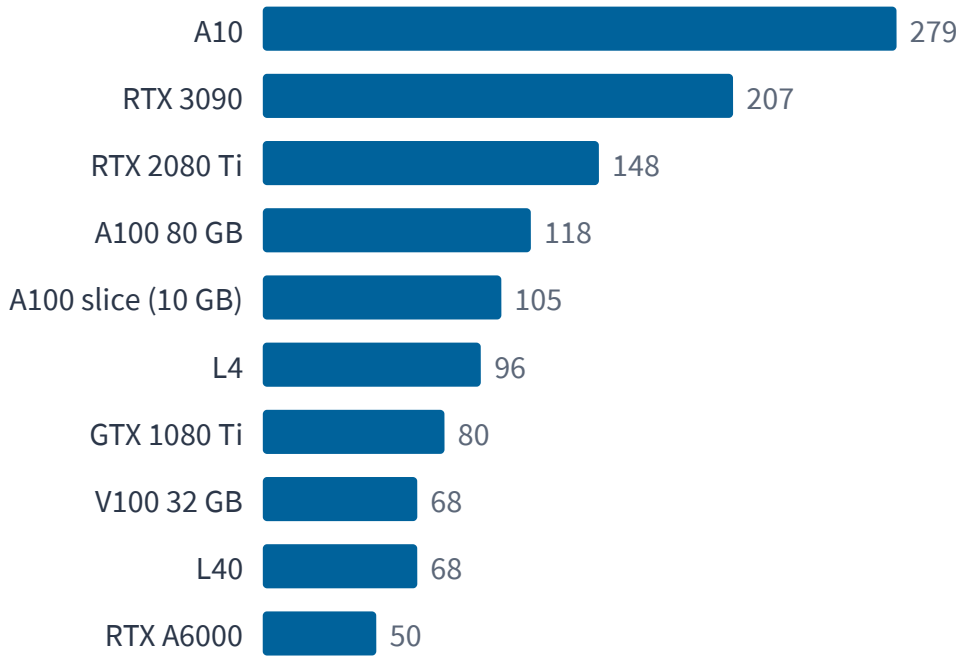
CPUs
about 35 thousand; most coursework needs nothing else

GPUs
1,460 in 33 models, from classroom cards to A100s

Storage
a home folder per student, shared course data, S3

AI models
hosted open-weight LLMs behind an OpenAI-style API

GPUs on NRP, the ten most common models



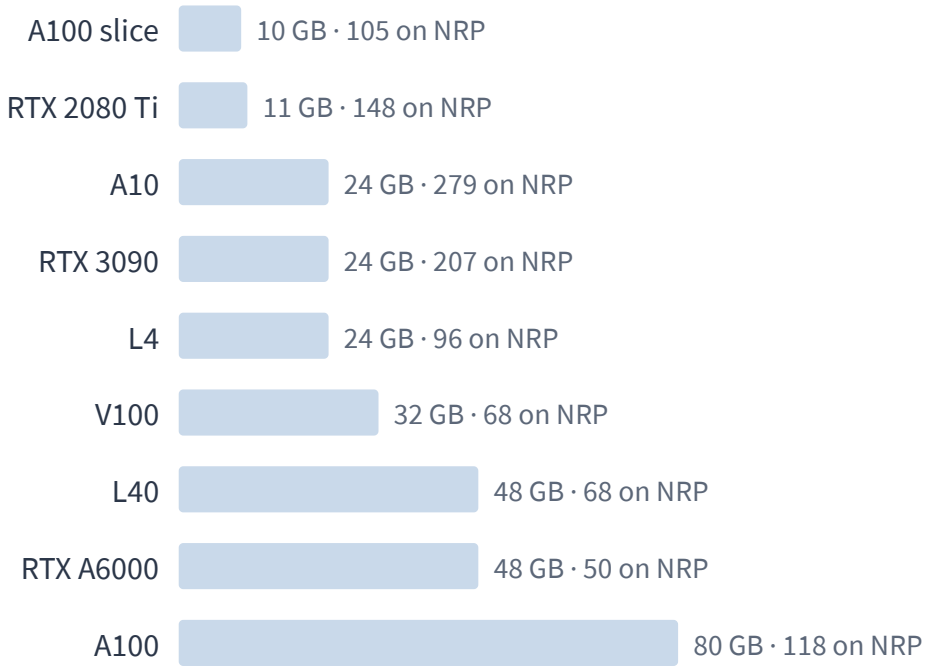
Cluster snapshot, 10 Nov 2026. Live view: nrp.ai/viz/resources

Ask for the kind of hardware the work needs, not the biggest card: most of a semester runs on CPUs.

Choosing GPUs for a Course

Intro data science: pandas, scikit-learn
CPUs only: 2 CPUs, 8 GB

GPU memory decides what fits

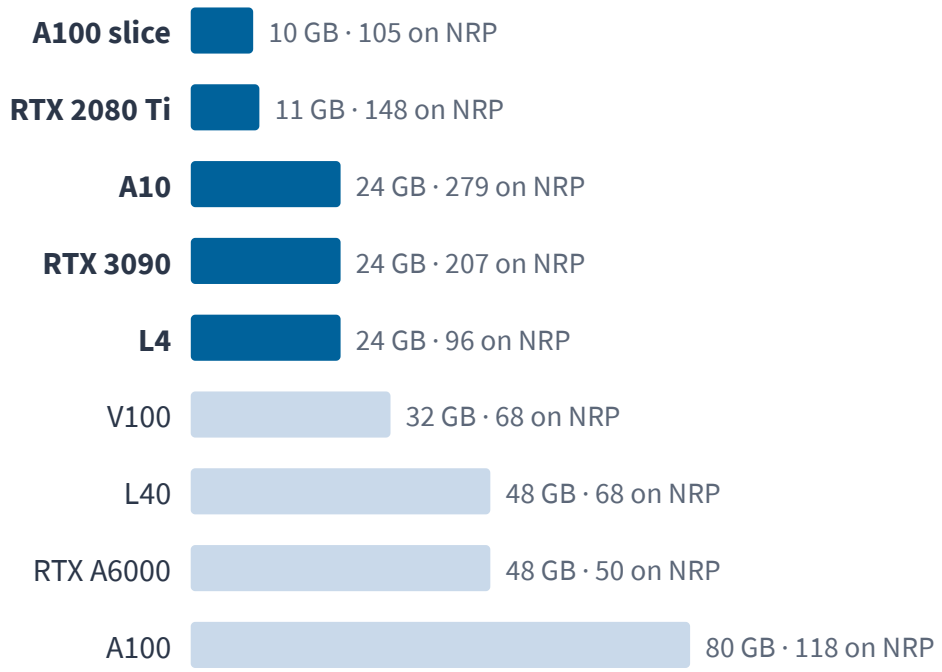


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one GPU with 10–24 GB

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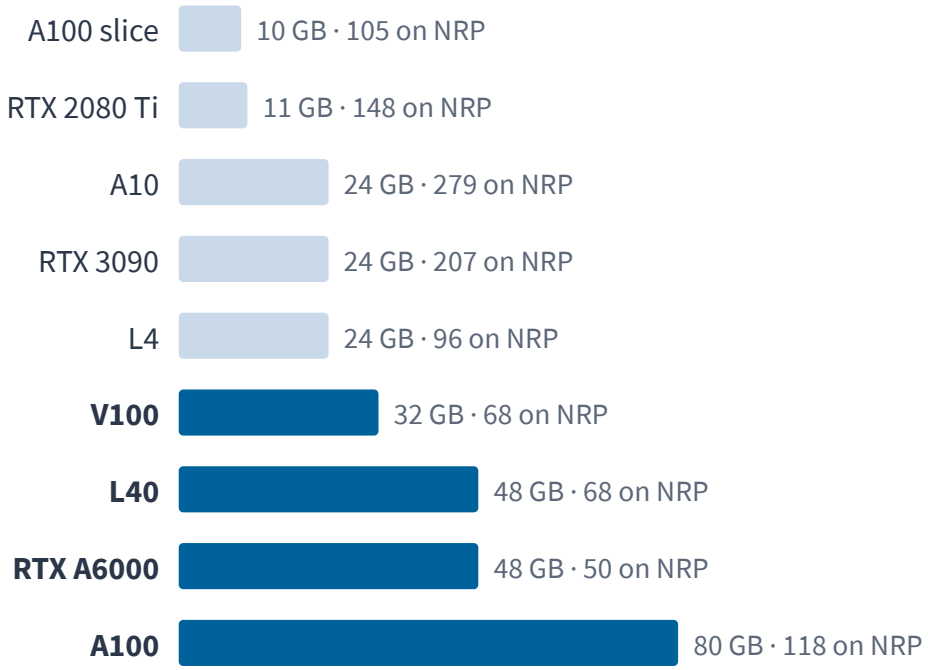
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Bigger models: fine-tuning, long training runs
48–80 GB, fewer students at once

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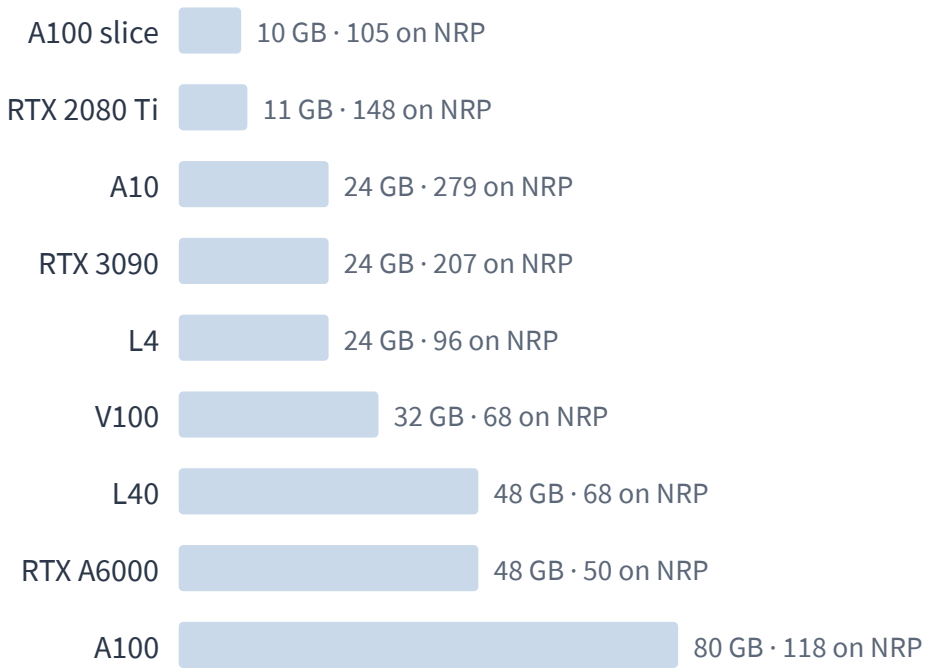
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Using LLMs, not training them
no GPU: NRP's hosted models

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Sharing GPUs Fairly

- **Reserved nodes:** on request, admins taint a contributor's nodes; only pods that **tolerate** the taint, the owner's, run there
  - Today: 119 nodes reserved for 23 groups



A pod without the toleration can't land on the reserved node, even on an idle GPU.

Sharing GPUs Fairly

- **Reserved nodes:** on request, admins taint a contributor's nodes; only pods that **tolerate** the taint, the owner's, run there
  - Today: 119 nodes reserved for 23 groups
- **Shared nodes:** idle GPUs go to **opportunistic** pods, the lowest priority class, e.g. OSG jobs for IceCube and LIGO
  - 232 opportunistic pods running, 208 of them from OSG



On a shared node, opportunistic pods fill the idle GPUs.

Sharing GPUs Fairly

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  - 232 opportunistic pods running, 208 of them from OSG
- When anyone needs that GPU, the opportunistic pod is **preempted**: it must checkpoint, or just restart

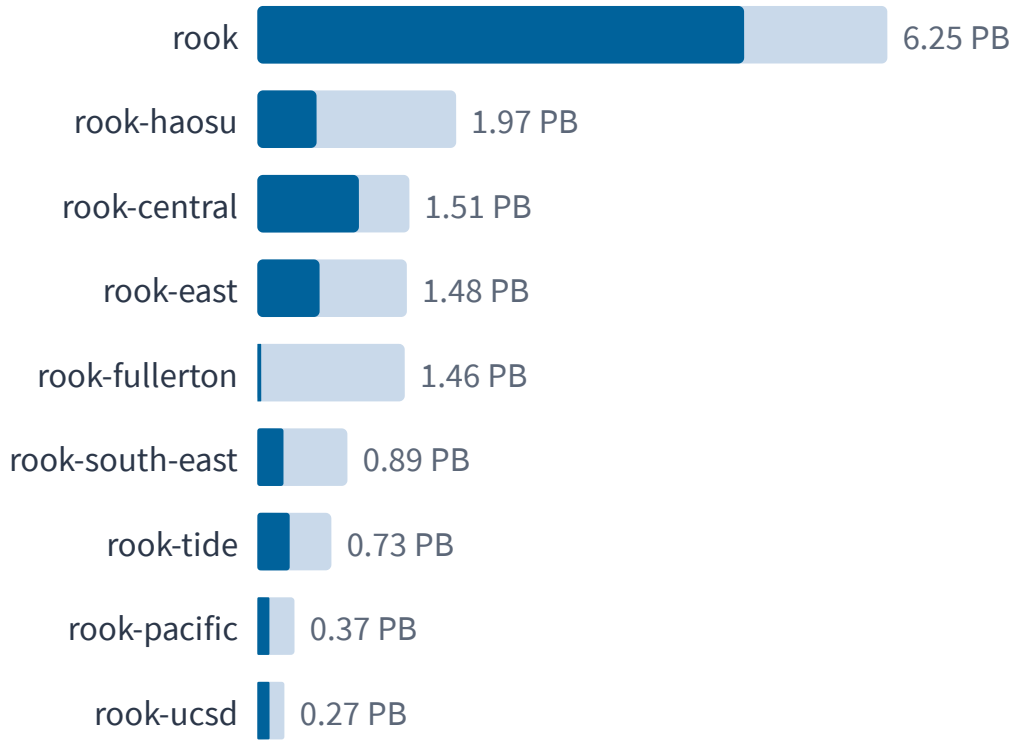


A normal pod arrives: the scheduler preempts an opportunistic pod to make room.

Regions and Storage

- Every node is labeled with its **region**, **zone** (143 zones) and **site**; workloads spread out or stay near their data
- **Rook** runs 9 **Ceph** clusters, one per region or large site: 14.9 PB raw
- CephFS for shared files, RBD for block, S3 for objects
- **Linstor** (DRBD) for fast replicated block storage; **CVMFS** for software; local NVMe for scratch
- Pick the storage in the region where you compute

Ceph clusters: raw capacity, and used



From the CephCluster resources, 10 Nov 2026

Ways to Run Work

Interactive: JupyterHub

- Pick a GPU from the start menu; explore, debug, teach
- Idle servers stop: 1 hour after you close the browser on NRP's hub, 6 hours at most on any hub

Interactive: a pod

- Start a container with kubectl, then open a shell in it
- Deleted after 6 hours; at most 2 GPUs

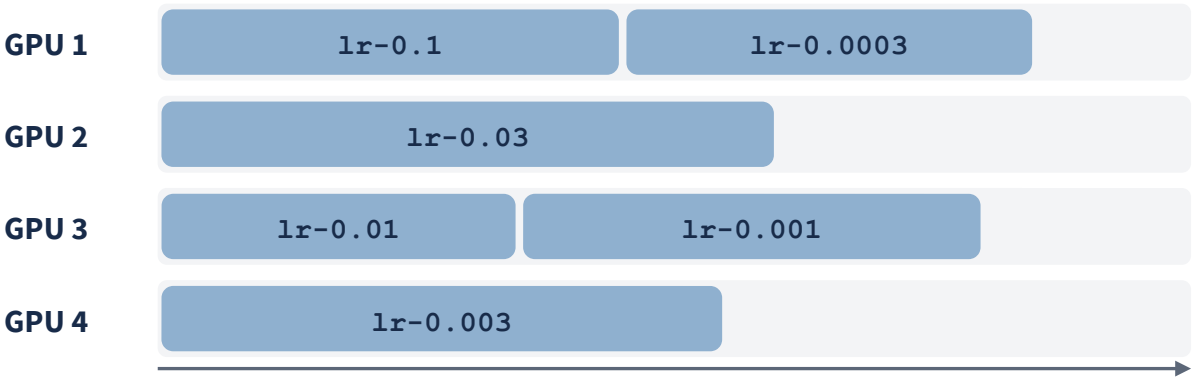
Long-running: a Deployment

- Keeps a service up (a web app, a database) and restarts it if it crashes
- Removed after 2 weeks unless admins grant an exception; no GPU if it sits mostly idle

Batch: a Job

- Runs your script to the end, then gives the GPU back
- Submit many at once; the rest wait in Pending for a free GPU

```
$ kubectl apply -f sweep/        # six training jobs
```



Six jobs on four GPUs: two wait in Pending and take the first GPUs that free up. When the last one ends, all four GPUs are free again.

A GPU Job

```
apiVersion: batch/v1
kind: Job
metadata: {name: gpu-train-daniel, namespace: my-lab}     # (1)
spec:
  template:
    spec:
      restartPolicy: Never
      containers:
      - name: train
        image: pytorch/pytorch:2.9.0-cuda13.0-cudnn9-runtime  # (2)
        command: ["python", "train.py"]
        resources:                                          # (3)
          requests: {cpu: "2", memory: 6Gi, nvidia.com/gpu: 1}
          limits:   {cpu: "2", memory: 6Gi, nvidia.com/gpu: 1}
      affinity:                                             # (4)
        nodeAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            nodeSelectorTerms:
            - matchExpressions:
              - {key: nvidia.com/gpu.memory, operator: Gt, values: ["20000"]}
              - {key: nvidia.com/gpu.compute.major, operator: Gt, values: ["7"]}
```

(1) **Where:** your namespace

(2) **What:** a container image, and the command to run

(3) **Resources:** 2 CPUs, 6 GB of memory, one GPU

(4) **Which GPU:** more than 20 GB, Ampere or newer; or name exact models

Using GPUs Fairly

Do

- Give GPUs only to the menu options that need them
- Let idle servers stop: keep the culling on
- Request what students will use; keep limits close to requests
- Keep data in home folders or the shared volume, not inside the image

Don't

- Leave servers holding GPUs overnight
- Give every student 4 GPUs “just in case”: use must stay above 40%
- Put results on the server's own disk: it vanishes with the server

https://nrp.ai/documentation/userdocs/start/policies/

JupyterHub for Your Class

Two Ways to Give Students Jupyter

NRP's JupyterHub

`jupyterhub-west.nrp-nautilus.io`

- Nothing to run: share the link
- Students choose CPUs, memory and a GPU from a menu
- NRP's images, the same rules for everyone
- Good for trying things, small classes and research

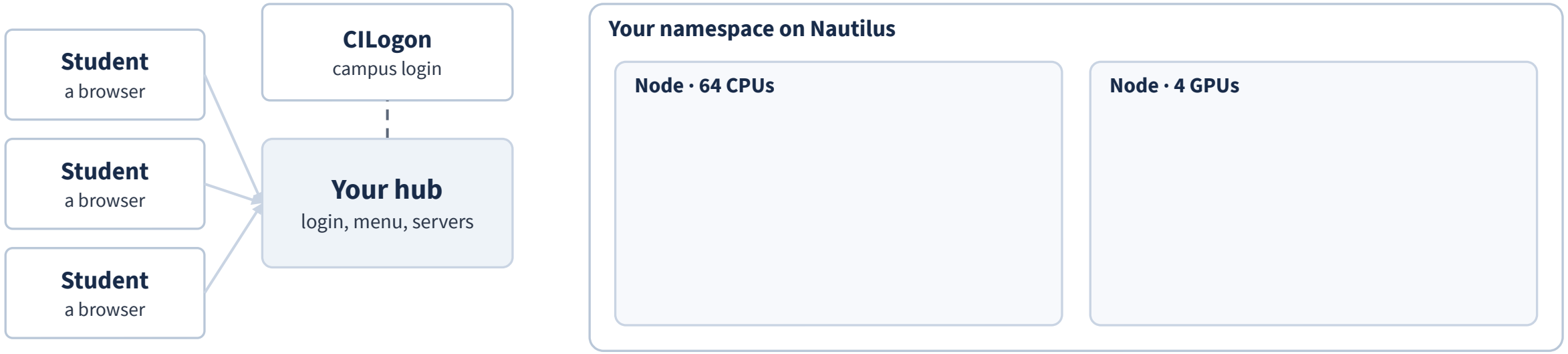
Your own hub, in your namespace

`jhub-yourcourse.nrp-nautilus.io`

- Your image menu, built for the syllabus
- Your login rules: only your students
- Your limits, storage and course files
- One Helm chart and one settings file: the rest of this section

152 JupyterHubs were running on NRP on 12 Nov, with 111 servers open at that moment.

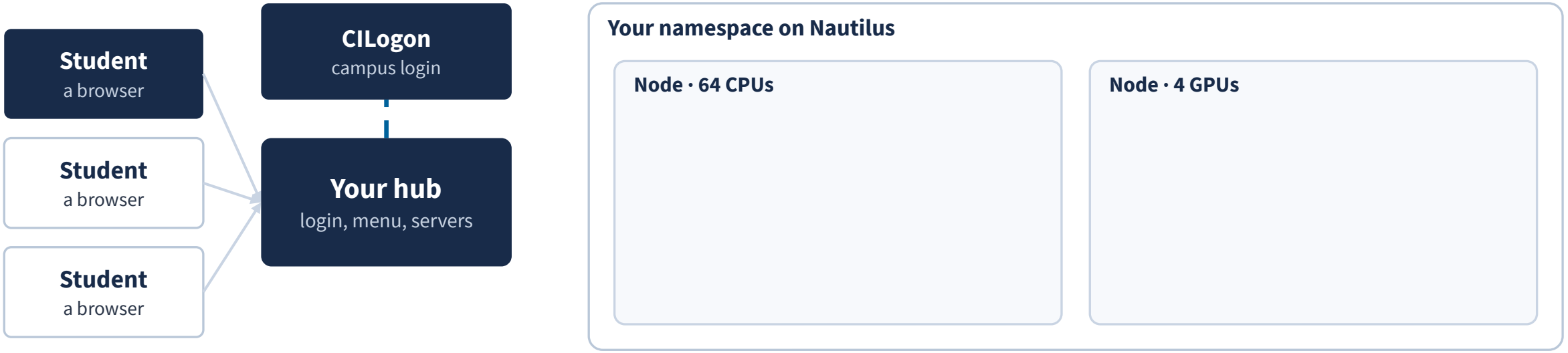
How a Class Hub Works



A class hub, step by step

One URL for the whole class. Behind it: your hub, in your namespace, and the cluster's nodes.

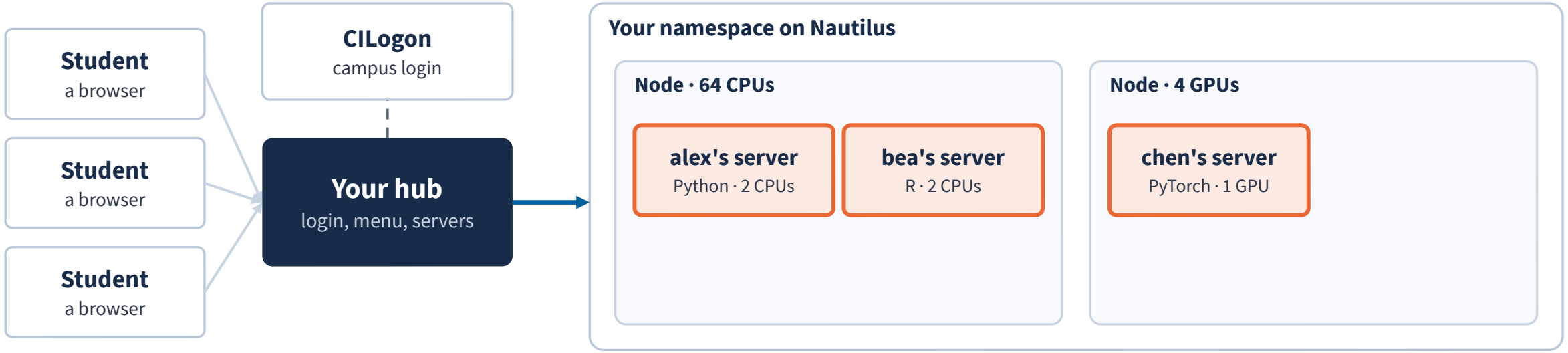
How a Class Hub Works



1. Students log in

With their campus account, through CILogon. The hub lets in only the people you allow.

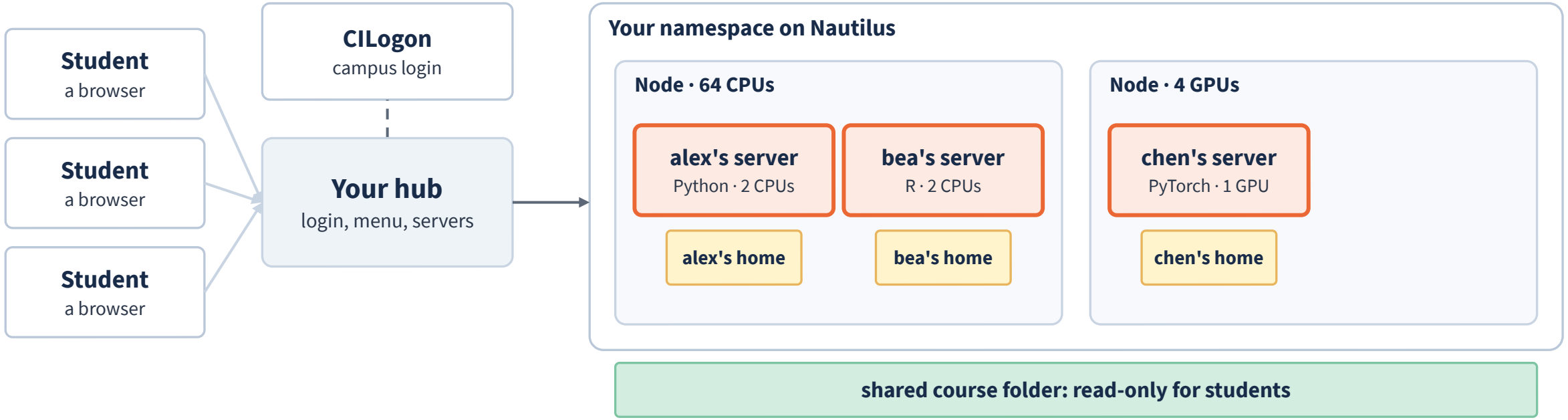
How a Class Hub Works



2. Each student gets a server

The hub starts a container for each student, with the image, CPUs, memory or GPU of the option they picked.

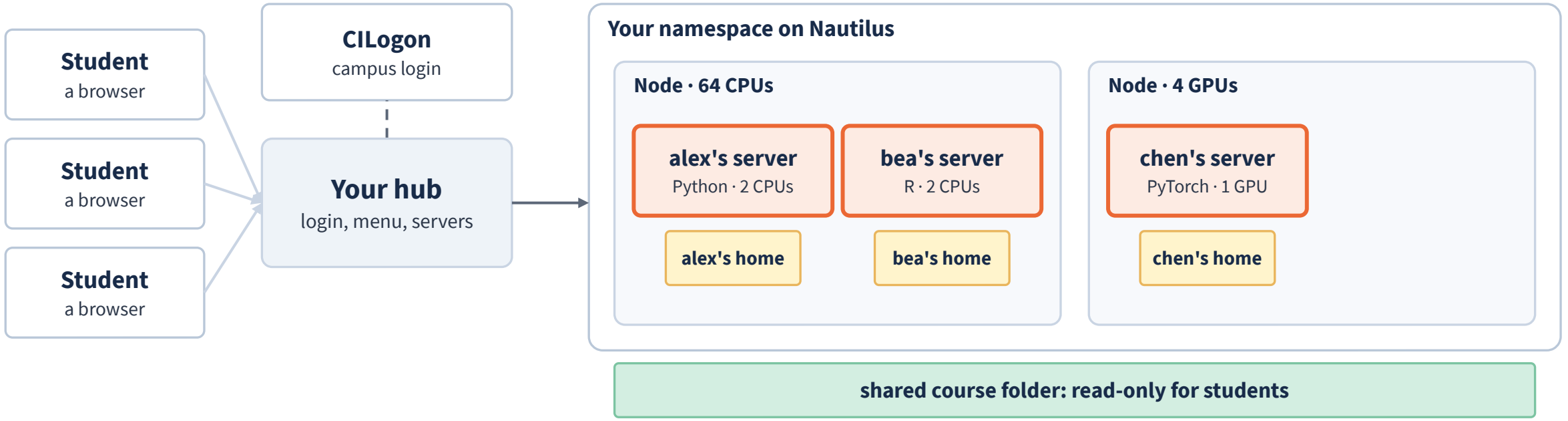
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3. Their files are there

Each server mounts the student's own home folder, plus a shared course folder everyone reads from.

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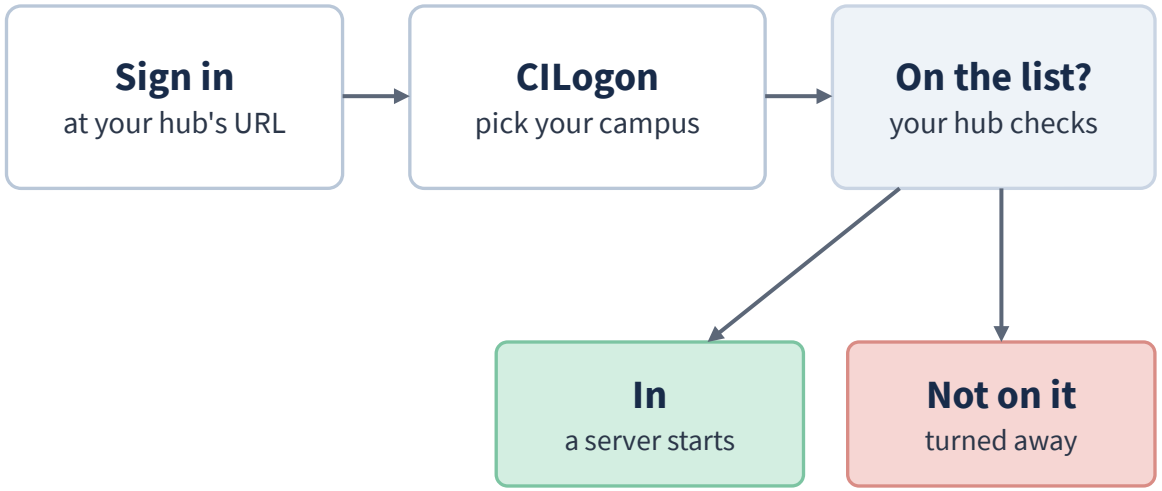


4. Idle servers stop

After an hour with no activity the server shuts down and its GPU goes back to the pool. The home folder stays for next time.

Who Can Log In

- **Campus login through CILogon:** you never hand out or reset passwords
- **Your class list:** allow your students by name or email, or a whole campus domain
- **Admins:** you and your TAs; an admin can open a student's server to help
- For a workshop or a demo, a shared password works too: today's demo uses one



CILogon connects your hub to your institution's own sign-in, the same one students use for email.

Setting Up Campus Login

The login part of values.yaml

```
hub:
  config:
    JupyterHub:
      authenticator_class: cilogon
    CILogonOAuthenticator:
      client_id: <from CILogon>        # and its client secret
      oauth_callback_url: https://jhub-bio210.nrp-nautilus.io/hub/oauth_callback
      allowed_idps:
        https://idp.campus.edu/idp/shibboleth:     # your campus's entity ID
          username_derivation: {username_claim: email}
      admin_users: [you@campus.edu, ta@campus.edu]
      allowed_users: [student1@campus.edu, student2@campus.edu]
```

1. Ask CILogon for a client
for your hub's address; it's issued by hand, so ask early

2. Look up your campus
its entity ID, in the CILogon IdP list

3. Name who gets in
admins, and your class list or your whole campus domain

4. helm upgrade
students who drop lose access at the next upgrade

Allow rules add up: `allowed_domains` admits a whole campus, `allowed_users` named people; `blocked_users` wins over both.

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          username_derivation: {username_claim: email}
      admin_users: [you@campus.edu, ta@campus.edu]
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          username_derivation: {username_claim: email}
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      allowed_users: [student1@campus.edu, student2@campus.edu]
```

1. Ask CILogon for a client
for your hub's address; it's issued by hand, so ask early

2. Look up your campus
its entity ID, in the CILogon IdP list

3. Name who gets in
admins, and your class list or your whole campus domain

4. helm upgrade
students who drop lose access at the next upgrade

Allow rules add up: `allowed_domains` admits a whole campus, `allowed_users` named people; `blocked_users` wins over both.

Your Image Menu

Server Options

Python for data science
2 CPUs · 8 GB

R with the tidyverse
2 CPUs · 8 GB

PyTorch with a GPU
4 CPUs · 16 GB · 1 GPU

BIO 210 course image
your packages, preinstalled

What students see when they start a server: a menu **you** write.

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Your own course image: exactly your packages (next slide)

Every student runs the same environment: no more "it works on my laptop".

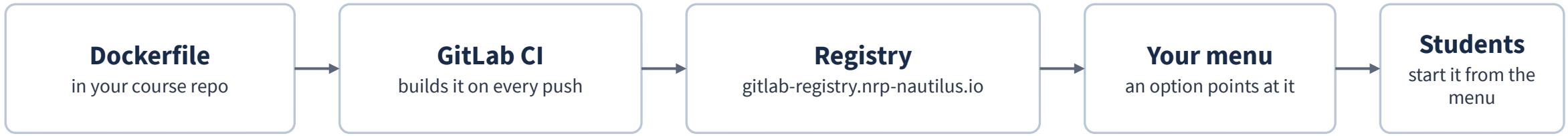
Your Course Image

Dockerfile

```
FROM quay.io/jupyter/scipy-notebook:2024-04-22

# what BIO 210 needs, installed once for everyone
RUN pip install --no-cache-dir \
      biopython scanpy seaborn statsmodels
```

- Start from a ready-made image; add only what your course needs
- NRP's GitLab builds it for you: no Docker on your laptop
- Pull it in class from the cluster's own registry: fast



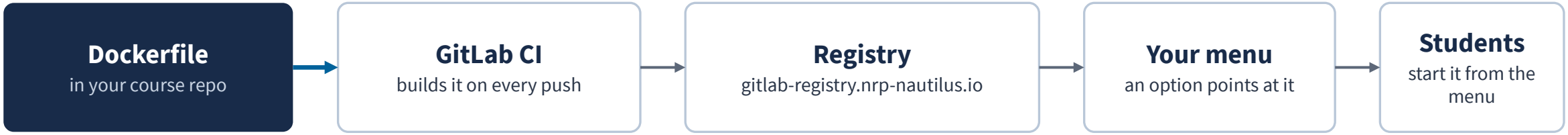
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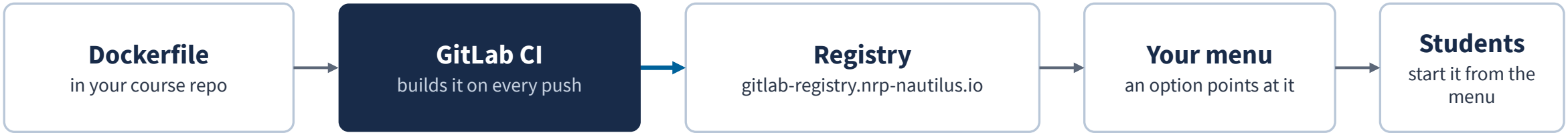
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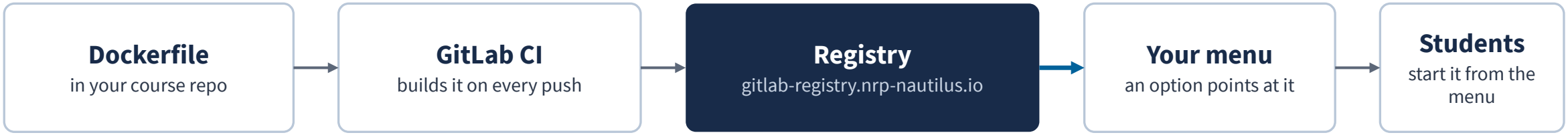
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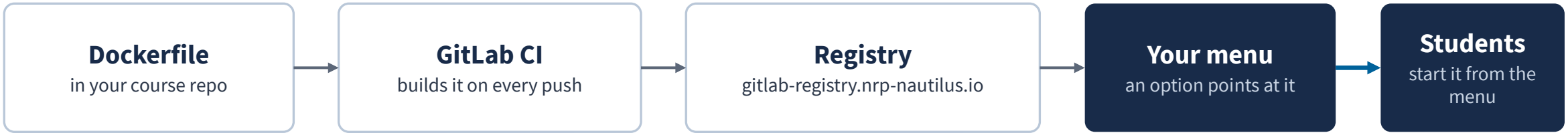
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```

- Start from a ready-made image; add only what your course needs
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Change a package, push, and next week's servers start with it.

Files and Course Material

Home folders

One per student, kept between sessions. You set the size; the demo uses 5 GB.

Course material by link

One link pulls your course's git repository into each student's folder (nbgitpuller). This webinar's launch links work the same way.

Shared data

One shared volume in every server: read-only for students, writable for you. Big datasets live once, not 40 times.

```
https://jhub-bio210.nrp-nautilus.io/hub/user-redirect/git-pull?repo=https://github.com/you/bio210&urlpath=lab/tree/bio210/week3.ipynb
```

A student clicks it and lands in week 3's notebook; their own edits are kept when the repository updates.

A Student's First Day

1. Click the link
from the syllabus or the LMS

A Student's First Day

1. Click the link
from the syllabus or the LMS

→

2. Campus login
CILogon, nothing new to remember

From the student's side, the first day is four clicks. No installs, no “which Python?”, and the same environment as everyone else.

Screenshots: NRP's own hub. Your hub's menu shows your options.

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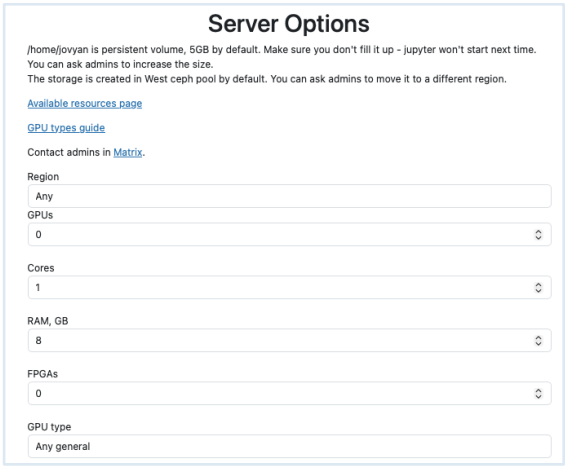
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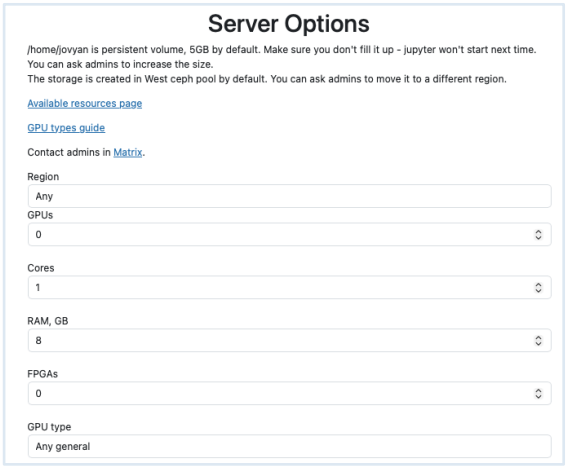
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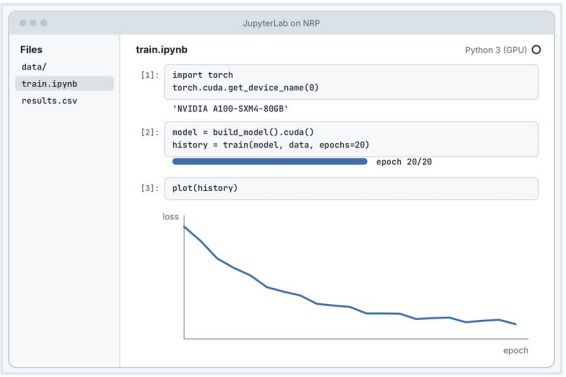
→

4. JupyterLab opens
week 1's notebook, ready to run

From the student's side, the first day is four clicks. No installs, no “which Python?”, and the same environment as everyone else.

Screenshots: NRP's own hub. Your hub's menu shows your options.



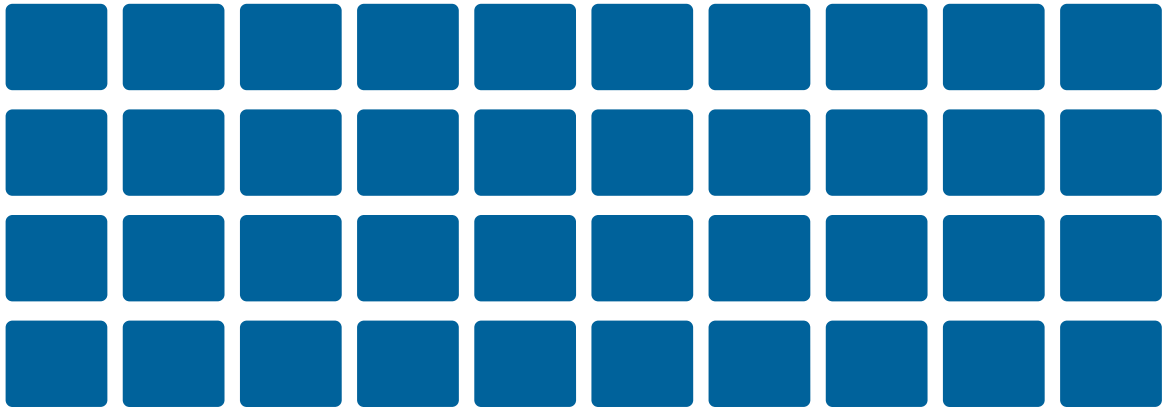


Sizing a Class

- Each server asks for the CPUs, memory and GPU of its menu option; the scheduler sets exactly that aside

Sizing a Class

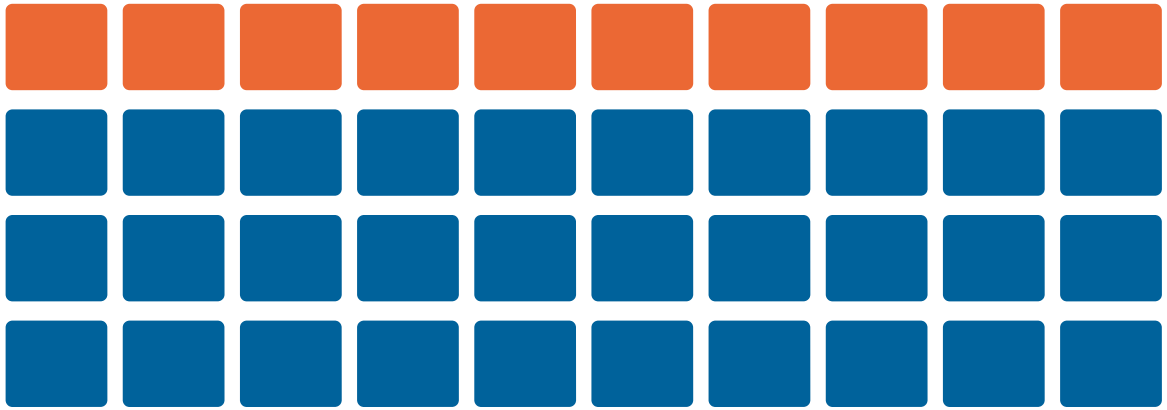
- Each server asks for the CPUs, memory and GPU of its menu option; the scheduler sets exactly that aside
- So set options to what a student really uses, not the most you can get



40 students on the Python option: 2 CPUs · 8 GB each
= 80 CPUs · 320 GB

Sizing a Class

- Each server asks for the CPUs, memory and GPU of its menu option; the scheduler sets exactly that aside
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- GPU options for the units that need them; the rest of the term on CPUs

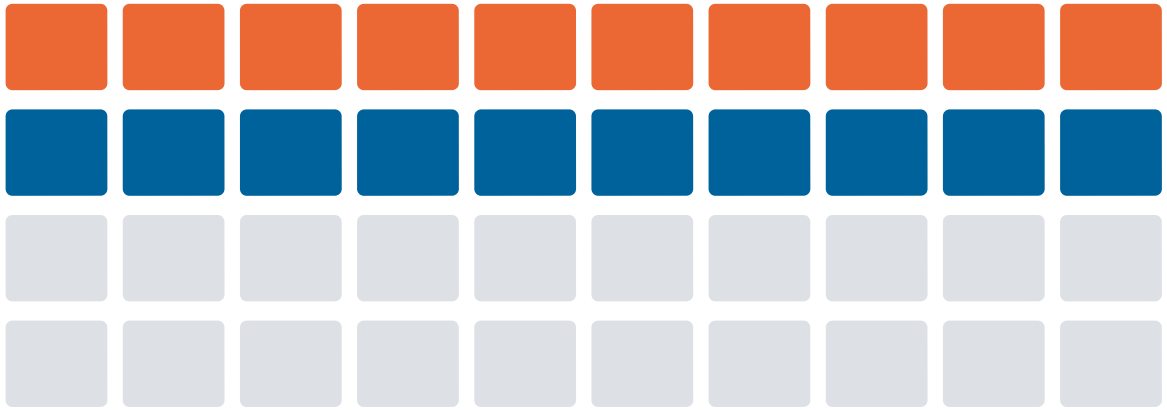


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10 students in the GPU lab: 1 GPU (24 GB) each
= 10 GPUs

Sizing a Class

- Each server asks for the CPUs, memory and GPU of its menu option; the scheduler sets exactly that aside
- So set options to what a student really uses, not the most you can get
- GPU options for the units that need them; the rest of the term on CPUs
- Idle servers stop, so a class only holds resources while it works
- Tell the NRP team your class size and your lab times



40 students on the Python option: 2 CPUs · 8 GB each
= 80 CPUs · 320 GB

10 students in the GPU lab: 1 GPU (24 GB) each
= 10 GPUs

20 servers idle after class: stopped after an hour
= resources back to the pool

Running the Class

Admin · BIO 210 hub

Student	Option	Server
alex@campus.edu	Python	running
bea@campus.edu	R	running
chen@campus.edu	PyTorch · GPU	stopped (idle)
dana@campus.edu	Python	running

“Open” drops you into the student's own JupyterLab, with their files, so you can see exactly what they see.

- The admin page lists every student's server: stop it, or **open it** to help
- Idle servers stop on their own: an hour by default, or what you set
- A big lab session? Ask the NRP team to set nodes aside for it
- End of term: students download their work, then you remove the hub

A Hub Is One File

values.yaml (shortened)

```
hub:
  config:
    JupyterHub:
      authenticator_class: cilogon          # campus login
      admin_users: [you@campus.edu, ta@campus.edu]
    CILogonOAuthenticator:
      allowed_idps:
        https://idp.campus.edu/idp/shibboleth:
          allowed_domains: [campus.edu]
singleuser:
  profileList:                              # the image menu
  - display_name: Python for data science
    kubespawner_override:
      image_spec: quay.io/jupyter/scipy-notebook:2024-04-22
  - display_name: PyTorch with a GPU
    kubespawner_override:
      image_spec: quay.io/jupyter/pytorch-notebook:cuda12-2024-04-22
      extra_resource_limits: {nvidia.com/gpu: "1"}
  storage:
    capacity: 5Gi                           # each student's home folder
cull:
  timeout: 3600                             # stop idle servers after an hour
ingress:
  hosts: [jhub-bio210.nrp-nautilus.io]      # your class's URL
```

Who logs in
campus login, your admins, your campus

The image menu
an image and resources per option

Home folders
the size of each student's folder

Idle servers
stop after an hour

The address
the URL you give the class

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the size of each student's folder

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stop after an hour

The address
the URL you give the class

```
helm install bio210 \
    jupyterhub/jupyterhub -f values.yaml
```

Changing the Hub Mid-Term

Week 1
`values.yaml`

the hub you agreed to run

Changing the Hub Mid-Term

Week 1
`values.yaml`

the hub you agreed to run

Week 6
`+ gpu-unit.yaml`

a GPU option for the deep-learning unit

```
helm upgrade bio210 jupyterhub/jupyterhub \
    -f values.yaml -f gpu-unit.yaml
```

Each change is a small file on top of the one you started with. Remove the file and upgrade again, and the change is gone; the base file never changes.

Changing the Hub Mid-Term

Week 1
`values.yaml`

the hub you agreed to run

Week 6
`+ gpu-unit.yaml`

a GPU option for the deep-learning unit

Week 9
`+ dataset.yaml`

this term's dataset, shared

```
helm upgrade bio210 jupyterhub/jupyterhub \
    -f values.yaml -f gpu-unit.yaml
```

Each change is a small file on top of the one you started with. Remove the file and upgrade again, and the change is gone; the base file never changes.

Changing the Hub Mid-Term

Week 1	Week 6	Week 9	Finals
`values.yaml`	`+ gpu-unit.yaml`	`+ dataset.yaml`	`+ finals.yaml`
the hub you agreed to run	a GPU option for the deep-learning unit	this term's dataset, shared	a longer idle timeout

```
helm upgrade bio210 jupyterhub/jupyterhub \
    -f values.yaml -f gpu-unit.yaml
```

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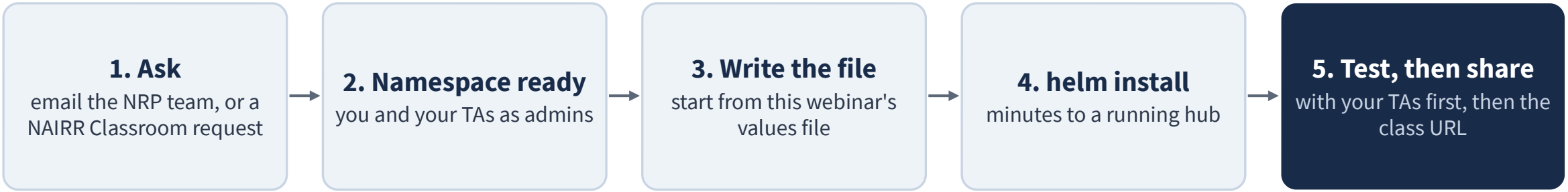
Changing the Hub Mid-Term

Week 1	Week 6	Week 9	Finals	After
`values.yaml`	`+ gpu-unit.yaml`	`+ dataset.yaml`	`+ finals.yaml`	`helm uninstall`
the hub you agreed to run	a GPU option for the deep-learning unit	this term's dataset, shared	a longer idle timeout	students keep their work

```
helm upgrade bio210 jupyterhub/jupyterhub \
    -f values.yaml -f gpu-unit.yaml
```

Each change is a small file on top of the one you started with. Remove the file and upgrade again, and the change is gone; the base file never changes.

From Request to First Lecture



Before the term: a week or two is plenty. The NRP team helps at every step, in office hours and on chat.

After the term: students download their work; `helm uninstall` removes the hub.

https://nrp.ai/documentation/userdocs/jupyter/jupyterhub/

This Webinar Runs on One

- **jh-training.nrp-nautilus.io**: a JupyterHub on NRP for workshops like this one
- Every attendee's server comes with `kubectl` and `helm` set up, and an LLM token
- Jupyter AI is in the image, pointed at NRP's models
- A pool of 100 namespaces, **nrp-training-000** to **099**: each person claims one for the session
- Sized for about 200 people at once; idle servers stop on their own

Workshop hub vs. course hub

Same chart, same kind of values file

A workshop: short-lived, a shared login, tools preinstalled

A course: a term long, campus login, your class list, home folders that last

If you follow the demo, you're using it right now.

AI in the Classroom

Hosted Models in JupyterLab

- NRP hosts open-weight models (Qwen, DeepSeek, Gemma, gpt-oss and more) behind one OpenAI-compatible endpoint
- **Jupyter AI** in JupyterLab: a chat panel, and an `%%ai` cell magic, pointed at NRP's models
- Put it in your course image: students log in and use it, no token handoff
- Or students call the models from Python, with a token from **nrp.ai/llmtoken**
- Free for research and education, like the rest of NRP

Notebook cells

```
%load_ext jupyter_ai_magics

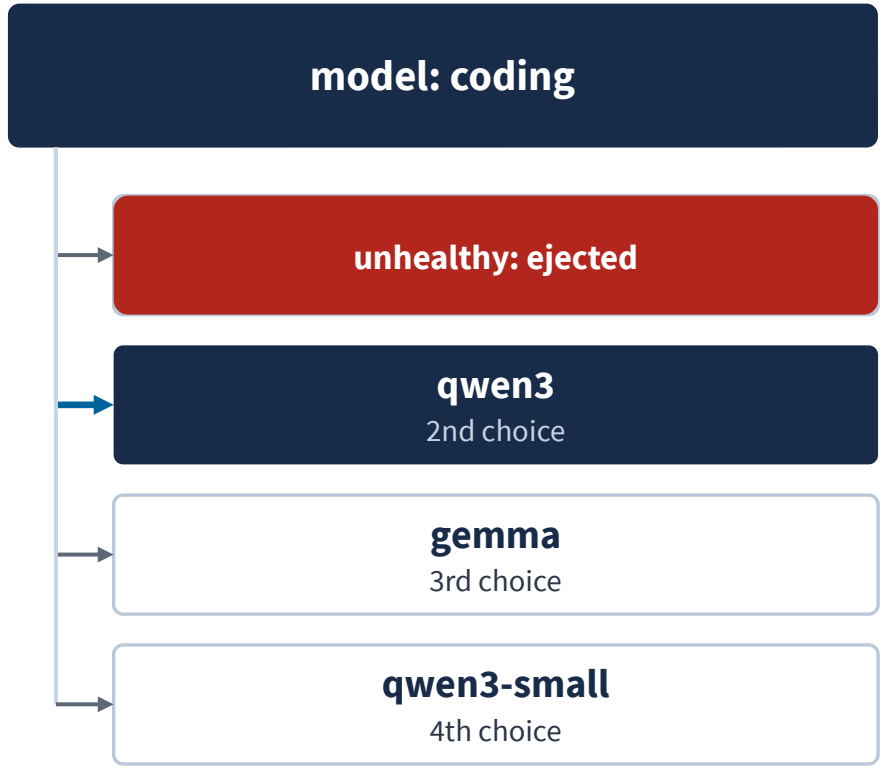
%%ai openai-chat:qwen3
Explain a p-value to a biology student in three sentences.

# or from Python, with a token from nrp.ai/llmtoken
from openai import OpenAI
client = OpenAI(base_url="https://ellm.nrp-nautilus.io/v1",
                api_key=os.environ["OPENAI_API_KEY"])
```

https://nrp.ai/documentation/userdocs/ai/llm-managed/

Models and Task Names

Ask for	Tried in this order
agents	kimi → deepseek-v4-flash → qwen3 → qwen3-small
coding	deepseek-v4-flash → qwen3 → gemma → qwen3-small
general	deepseek-v4-flash → qwen3 → qwen3-small → minimax
reasoning	deepseek-v4-flash → minimax → qwen3-small → qwen3
vision	qwen3 → qwen3-small → kimi → deepseek-v4-flash
long-context	deepseek-v4-flash → gemma → kimi → minimax
kubernetes	kimi → gemma → qwen3 → qwen3-small
cuda	deepseek-v4-flash → qwen3-small → gpt-oss → gemma-small-e4b
service	deepseek-v4-flash → minimax → qwen3 → qwen3-small → gemma → gemma-small → gpt-oss



Only when it's down, or ejected by the health checks, does the 2nd choice answer.

Live Demo

Getting Help

- **Office hours** with the NRP team, every two weeks
- **Chat** with the team and other instructors, for quick questions
- **Docs** for every step in today's talk, at nrp.ai/documentation
- **Recorded trainings** on Docker, Kubernetes and JupyterHub, at nrp.ai/training
- **Today's materials**: the slides, the demo notebook and the values files

Bringing a course?

Email the NRP team with the course, the number of students and the start date, and they'll have your namespace ready before the first lecture.

https://nrp.ai/contact/

Where to Start

This term

- Log in at **nrp.ai** and try NRP's JupyterHub
- Email the NRP team about your course: what it is, how many students, when it starts
- For priority, a **NAIRR Classroom** request
- Come to office hours: every two weeks, with the NRP team

Your own hub

- Start from today's values file: it's in the materials
- Add CILogon and your class list before students arrive
- Build a course image when the default menu isn't enough
- Follow along later: every step of today's demo is on the training page

https://nrp.ai/education/

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- The NRP operations teams at SDSC, the University of Nebraska–Lincoln and MGHPCC, and the campuses that host and share their hardware
- Forough Ghahramani and Edge, for hosting this webinar series

Questions?

Today's materials	`training.nrp-nautilus.io/nairr-classroom`
Portal and login	`nrp.ai`
Teaching with NRP	`nrp.ai/education`
Request a namespace	`get-started.nrp-nautilus.io`
NRP's JupyterHub	`jupyterhub-west.nrp-nautilus.io`
Your own hub (docs)	`nrp.ai/documentation/userdocs/jupyter/jupyterhub`
LLM token	`nrp.ai/llmtoken`
Support chat and email	`nrp.ai/contact`

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Backup

NRP's Hub or Your Own?

	NRP's JupyterHub	Your own hub
Setup	none: share the link	one values file, helm install
Who logs in	anyone with an NRP login	your class list, through CILogon
Images	NRP's menu	your menu, or your course image
Resources	students choose from the menu	you set them per option
Course files	students copy them in	a link, and a shared folder
Help	NRP's support chat	you open a student's server